CHAPTER 1

INTRODUCTION

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In the aviation industry, pilots are required to maintain several critical records such as medical certificates, flight logbooks, licenses, and personal information to comply with regulatory and safety standards. Traditionally, these records are maintained manually or stored across different platforms, which often leads to difficulties in data management, reduced accessibility, and increased risk of data loss or unauthorized access. As aviation operations become more digital, there is a growing need for a secure and centralized system to manage pilot records efficiently.

1.1 OVERVIEW

This project presents a secure web-based aviation record management system designed to centralize and digitalize important pilot records such as medical certificates, licenses, flight logbooks, and personal details. Traditionally, these records are maintained manually or stored across multiple platforms, which can lead to difficulties in tracking, reduced accessibility, and potential security risks. The proposed system addresses these challenges by providing a unified digital platform developed using the MEAN stack, which includes MongoDB, Express.js, Angular, and Node.js. The system ensures secure access through JSON Web Token (JWT) based authentication, allowing only authorized users to view and manage sensitive aviation data. It supports complete CRUD operations for all modules and includes a dashboard that displays record summaries, expiry alerts, and recent updates.

1.2 RETRIEVAL-BASED CHATBOT

In the project titled "A Secure Web-Based Aviation Record Management System with Integrated Retrieval-Based Chatbot Using MEAN Stack," the retrieval-based chatbot plays a central role in enabling intelligent and user-friendly access to pilot records. Unlike traditional navigation methods that require users to manually browse different modules, the chatbot allows pilots to retrieve information using natural language queries. The chatbot is designed as a lightweight retrieval system rather than a complex machine learning or deep learning model, making it suitable for institutional deployment with limited computational resources. The chatbot operates on a keyword-based intent mapping mechanism. When a user submits a query, the system analyzes the input text and identifies relevant keywords such as "medical," "license," "logbook," or "profile." Based on the detected intent, the chatbot maps the query to the corresponding MongoDB collection. For example, queries containing terms like "Class 1" or "medical expiry" are directed to the medical records module, while queries related to "CPL," "PPL," or "license validity" are routed to the license collection. This structured mapping ensures accurate and efficient retrieval of information.

1.3 AVIATION RECORD MANAGEMENT SYSTEM

The project titled **"A Secure Web-Based Aviation Record Management System with Integrated Retrieval-Based Chatbot Using MEAN Stack"** provides an efficient and structured solution for managing pilot-related aviation records in a secure digital environment. The system focuses on organizing and maintaining essential records such as medical certificates, flight logbooks, pilot licenses, and personal information, which are critical for regulatory compliance.

1.4 PROBLEM STATEMENT

In the aviation industry, pilots are required to maintain multiple critical records such as medical certificates, flight logbooks, licenses, and personal information to comply with regulatory and safety requirements. Traditionally, these records are maintained manually or stored across different platforms, which often leads to challenges such as data fragmentation, difficulty in tracking document validity, limited accessibility, and increased risk of data loss or unauthorized access. Managing these records efficiently becomes more complex as the volume of data increases over time. Additionally, retrieving specific information from these records can be time-consuming, especially when pilots or aviation institutions need quick access to details such as medical expiry dates, license validity, or recent logbook entries. The absence of a centralized and secure digital system results in operational inefficiencies and increases the likelihood of missed renewals or compliance issues. Therefore, there is a strong need for a centralized digital Pilot Information Management System that can securely store, manage, and retrieve pilot records in an efficient manner. Such a system would help in maintaining organized data, providing timely alerts for document renewals, improving accessibility, and ensuring better compliance with aviation regulations while enhancing overall operational efficiency.

1.5 IMPORTANCE OF WORK

The development of a secure web-based aviation record management system offers significant advantages in improving operational efficiency, data security, and regulatory compliance within the aviation sector. By digitalizing and centralizing pilot records, the proposed system addresses several challenges faced by pilots and aviation institutions.

1) **Improved Record Accessibility:**
   The system enables pilots and authorized personnel to access important records such as medical certificates, licenses, and logbooks anytime and from anywhere, eliminating dependency on physical documents.

2) **Enhanced Data Security:**
   By implementing JWT-based authentication and secure backend services, the system ensures that sensitive aviation data is protected from unauthorized access, data breaches, and accidental loss.

3) **Efficient Compliance Management:**
   Automated tracking of expiry dates for medical and license documents helps prevent missed renewals and ensures continuous compliance with aviation regulations and safety standards.

4) **Time-Saving and User-Friendly Access:**
   The integration of a retrieval-based chatbot allows users to quickly retrieve required information through natural language queries, reducing manual navigation and saving time.

1.6 OBJECTIVES

- Develop a secure web-based system for managing aviation records.
- Implement JWT-based authentication to protect pilot data.
- Enable efficient storage, update, and retrieval of pilot records such as medicals, licenses, and logbooks.
- Integrate a retrieval-based chatbot for quick access to records using natural language queries.
- Improve efficiency and accessibility of aviation record management.

1.7 SOCIAL IMPACT

- Helps improve aviation safety by ensuring valid and up-to-date pilot records.
- Provides a reliable and accessible digital platform for pilots and aviation institutions.
- Reduces administrative workload by automating record management and retrieval.
- Enhances data security and reduces the risk of data loss or unauthorized access.
- Encourages digital transformation in the aviation and training sectors.

1.8 CHALLENGES

- Keeping pilot records safe and secure from unauthorized access.
- Making sure the chatbot gives correct and useful information for user queries.
- Tracking expiry dates of medical and license records accurately.
- Managing user login and authentication securely.
- Maintaining good system performance as the number of users increases.

1.9 LIMITATION

- The system depends on accurate and regularly updated aviation records to function effectively.
- The chatbot relies on predefined keywords and may not understand complex user queries.
- Data security and privacy risks may arise if proper authentication and encryption mechanisms are not implemented.
- Users who are not familiar with digital platforms may initially face difficulty while using the system.

1.10 DOCUMENT ORGANIZATION

Our project focuses on developing a Secure Web-Based Aviation Record Management System with an integrated retrieval-based chatbot in an efficient and secure manner.

- **Chapter 2** reviews existing systems and related work in aviation record management and web-based security solutions.
- **Chapter 3** presents the hardware and software requirements needed to implement the system.
- **Chapter 4** explains the system design, including the overall architecture and description of different modules such as profile, medical, license, logbook, and chatbot.
- **Chapter 5** describes the implementation details of each module and the integration of frontend, backend, and database components.
- **Chapter 6** presents the results and performance analysis of the developed system.
- **Chapter 7** discusses the conclusion and future enhancements of the project.
- **Chapter 8** explains the SDG mapping and PO mapping related to the project.

CHAPTER 2

LITERATURE SURVEY

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LITERATURE SURVEY

This section presents a review of various research works related to web-based record management systems, secure authentication mechanisms, and the use of chatbots for information retrieval. The surveyed studies focus on digital record management, cloud-based applications, security techniques such as JSON Web Tokens (JWT), and conversational interfaces used to improve user interaction. Studies also examine encryption techniques and secure communication protocols to protect data during transmission.

2.1 WEB-BASED RECORD MANAGEMENT SYSTEMS

Several researchers have proposed web-based systems to manage sensitive records in domains such as healthcare, education, and transportation. Smith et al. [11] developed a centralized web-based document management system using a client–server architecture to improve data accessibility and reduce manual record handling. The system emphasized role-based access control and secure data storage, demonstrating improved efficiency compared to traditional paper-based methods. The study highlighted the importance of centralized data repositories for maintaining accuracy and consistency across records. The system emphasized role-based access control and secure data storage, demonstrating improved efficiency compared to traditional paper-based methods. The study highlighted the importance of centralized data repositories for maintaining accuracy and consistency across records.

2.2 SECURE WEB-BASED AVIATION RECORD SYSTEMS AND RETRIEVAL MODELS

Several researchers have explored secure web-based systems and intelligent retrieval mechanisms for managing structured records in specialized domains. Sharma et al. [23] reviewed different web technologies for secure data management and emphasized the importance of scalable architectures and database optimization in handling large volumes of structured records. Their study demonstrated how centralized digital systems improve data consistency and reduce manual errors compared to traditional record-keeping methods. They also highlighted the significance of authentication mechanisms such as token-based security to protect sensitive information. In another study, Lee et al. analyzed RESTful API security using JSON Web Tokens (JWT) and compared it with session-based authentication models.

2.3 ECURE ARCHITECTURE AND DATA RETRIEVAL SYSTEMS

Lakshminarayanan et al. [5] proposed a secure web-based data management framework focusing on structured record storage and efficient retrieval mechanisms. Their system implemented modular architecture principles to manage user records through separate functional modules, improving scalability and maintainability. The study emphasized the importance of database schema design, indexing, and validation techniques to ensure data consistency and optimized performance. Various backend technologies were compared, and RESTful API architecture demonstrated better flexibility and integration capabilities.

2.4 EVALUATION USING RETRIEVAL-BASED CHATBOT

Murugan et al. [17] explored the application of machine learning algorithms for skin cancer prediction. In contrast, this project focuses on the use of a retrieval-based chatbot for aviation record management. The chatbot is designed to help users quickly access stored pilot information such as medical certificates, licenses, and logbook entries through natural-language queries. The system uses keyword-based intent detection to identify the type of information requested by the user. For example, queries containing words like “medical,” “license,” or “logbook” are mapped to the corresponding database modules. The chatbot then retrieves the relevant records from the MongoDB database and presents them in a readable format. The performance of the chatbot depends on accurate keyword matching, proper database design, and structured storage of records. Testing showed that the chatbot can efficiently handle common queries related to expiry dates, personal details, and recent flight activities. Compared to manual navigation through multiple pages, the chatbot significantly reduces the time required to locate specific information. The evaluation highlights that retrieval-based chatbots are lightweight, fast, and suitable for domain-specific applications such as aviation record management. Although advanced AI chatbots may provide more flexible conversations, the proposed approach ensures reliability, security, and low computational requirements, making it ideal for educational institutions and small-scale deployments.

2.5 AVIATION RECORD MANAGEMENT SYSTEMS

Several researchers have developed digital systems to manage aviation-related records such as pilot licenses, medical certificates, and flight logbooks. Traditional methods rely on manual documentation or fragmented software systems, which can lead to data loss, difficulty in tracking expiry dates, and reduced accessibility. Recent studies have focused on web-based solutions that centralize pilot information in secure databases. These systems improve efficiency by allowing users to store, update, and retrieve records from a single platform. The use of cloud databases ensures data availability, backup, and scalability. Some research also highlights the integration of conversational interfaces, such as chatbots, to simplify data access. Retrieval-based chatbots enable users to obtain specific information using natural-language queries instead of navigating through multiple screens. This approach reduces search time and improves user experience.

2.6 HIGH-EFFICIENCY DIGITAL MANAGEMENT SYSTEM

Patil, A., & Bhirud, S. et al. [19] discussed intelligent systems designed to improve accuracy and efficiency in data processing applications. In the context of aviation record management, modern web-based systems aim to securely store and retrieve pilot information with high reliabilit Patil, A., & Bhirud, S. et al. [19] discussed intelligent systems designed to improve accuracy and efficiency in data processing applications. In the context of aviation record management, modern web-based systems aim to securely store and retrieve pilot information with high reliability. Technologies like JWT authentication and cloud databases help maintain data integrity and security while supporting multiple users.

The proposed system manages various records such as medical certificates, licenses, personal details, and flight logbook entries within a single platform. By using a structured database and modular design, the system ensures accurate storage and quick retrieval of information without data loss. A key feature of the system is the retrieval-based chatbot, which enables users to obtain specific information through simple text queries. Instead of manually searching through multiple modules, users can directly ask for details such as document expiry dates or recent flight records. This significantly improves efficiency and reduces user effort. System performance depends on factors such as database organization, secure authentication, and efficient server processing.

2.7 METHODOLOGY FOR AVIATION RECORD MANAGEMENT SYSTEM

Andal Priya et al. [24] discussed various methods used in intelligent systems for accurate data processing and classification. In aviation record management, the focus is on efficient storage, retrieval, and monitoring of critical pilot documents rather than medical diagnosis. The proposed system uses a structured methodology to manage records such as medical certificates, licenses, personal details, and flight logbook entries. All data is stored in a centralized cloud database, ensuring easy access and reliability. Proper validation and organization of data improve accuracy and reduce the chances of errors. The system follows a modular approach, where each type of record is handled by a separate module. Users can create, view, update, and delete records through a secure interface. Authentication using JSON Web Tokens (JWT) ensures that only authorized users can access their data.

CHAPTER 3

REQUIREMENT SPECIFICATION

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REQUIREMENT SPECIFICATION

This chapter describes the hardware and software requirements needed to successfully develop and implement the proposed Secure Web-Based Aviation Record Management System with Integrated Retrieval-Based Chatbot using MEAN Stack. These requirements ensure smooth operation of the web application, secure data handling, and efficient performance of the chatbot system. The requirements are categorized into hardware and software specifications.

3.1 HARDWARE REQUIREMENTS

The following hardware is required to develop and run the proposed Aviation Record Management System with Chatbot using Docker. Since the application runs inside containers, a computer with moderate configuration is sufficient.

• **Operating System:** Windows 10 or above (64-bit) — required to run Docker Desktop.
• **Storage (SSD):** 256 GB or more — provides faster loading of Docker images and project files.
• **RAM:** Minimum 8 GB — required for running Docker containers smoothly.
• **Processor Speed:** 2.0 GHz or higher — ensures efficient execution of containers.
• **Processor:** Intel Core i5 / AMD Ryzen 5 or higher — recommended for Docker-based applications.
• **Virtualization Support:** Enabled in BIOS — required for Docker to run properly.
• **Internet Connection:** Required for downloading Docker images and access.

3.2 SOFTWARE REQUIREMENTS

The following software is needed to develop and run the proposed Aviation Record Management System with Chatbot using Docker.

- **Operating System:** Windows 10 or above — supports all required tools.
- **Docker Desktop:** Used to run the application in containers.
- **Node.js:** Runs the backend server.
- **Angular:** Used to create the frontend of the web application.
- **MongoDB Atlas:** Used to store all pilot data in the cloud database.
- **Express.js:** Handles server requests and APIs.
- **Web Browser:** Google Chrome / Edge / Firefox — used to open and use the system.
- **Visual Studio Code:** Used to write and edit the project code.

CHAPTER 4

SYSTEM DESIGN

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SYSTEM DESIGN

The proposed Secure Web-Based Aviation Record Management System with Integrated Chatbot is designed with a user-friendly interface that allows pilots to securely manage their aviation records. Users can log in to the system and add or view information such as personal details, medical certificates, licenses, and flight logbook entries. The system processes the input data by validating and storing it in a structured format within the MongoDB cloud database. Necessary checks such as authentication, data verification, and expiry monitoring are applied before saving the records. . When a user interacts with the chatbot, the system analyzes the query, identifies relevant keywords, and retrieves the required information from the database.

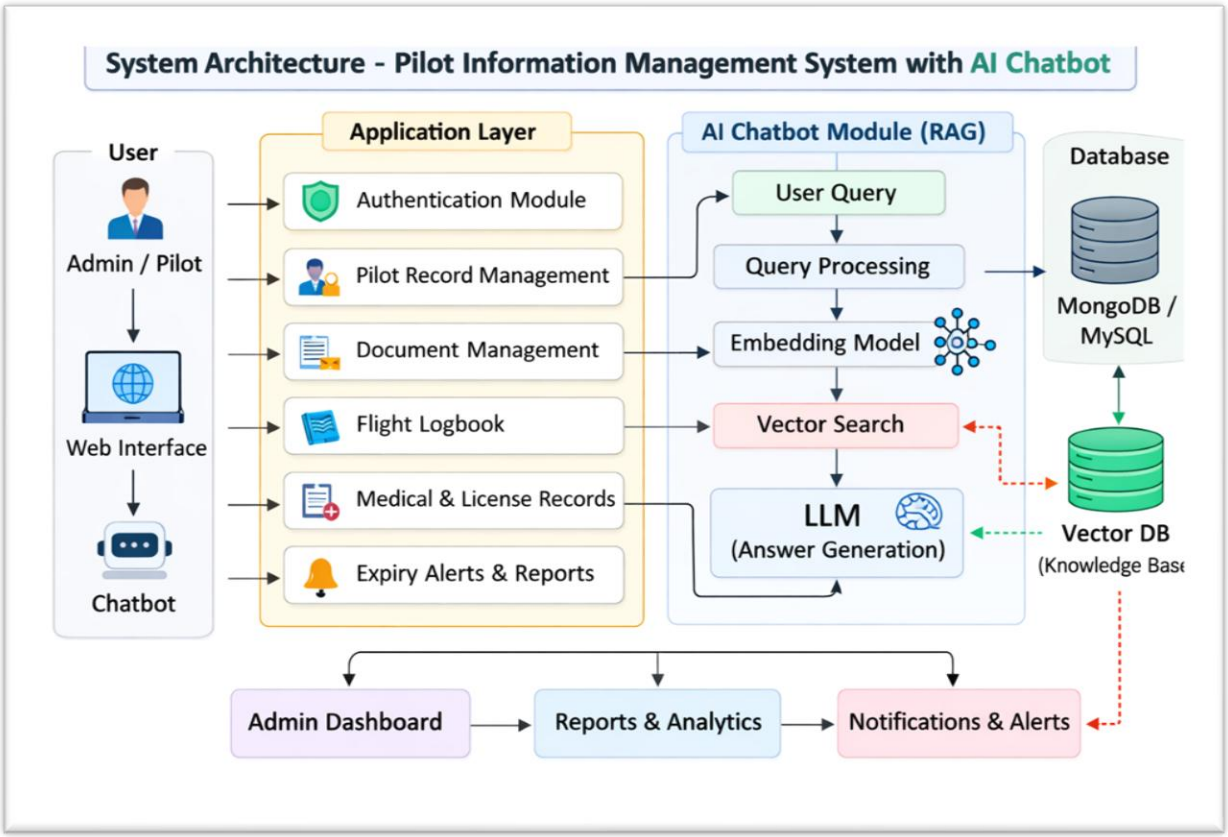

Fig 4.1 System design

4.1 DATA PREPROCESSING

Data preprocessing in this system means preparing user data before storing it in the database. The system handles pilot records such as personal details, medical certificates, licenses, and logbook entries.

- **Resize the images**: Make sure all images are the same size (e.g., 224x224 pixel).
- **Normalize pixel values**: Convert pixel values to a range of [0, 1] by dividing by 255.
- **Data augmentation**: Apply transformations like rotation, flipping, zooming, and shifting to increase the diversity of your training data.
- **Data Splitting**: Divide the dataset into training and testing sets, typically with an (80% / 20%) or (70% / 30%) split.

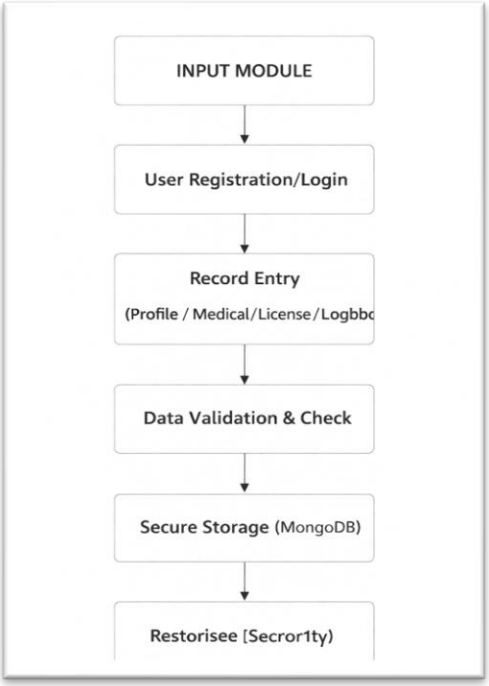

Fig 4.2 Input module

4.2 DATA SEGMENTATION

In the proposed system, data segmentation refers to organizing pilot records into different categories or modules. Instead of images, the system handles structured information such as personal details, medical certificates, licenses, and flight logbook entries.

4.3 FEATURE EXTRACTION

Feature extraction in this system refers to identifying important information from stored records that can be used for display, alerts, and chatbot responses.

• **Key Data Identification:** Important fields such as expiry dates, license numbers, flight hours, and contact details are extracted from the database.

• **Status Calculation:** The system determines whether documents are valid, expiring soon, or expired based on stored dates.

• **Dashboard Information:** Summary details such as recent logbook entries and alerts are generated from the extracted data.

4.3.1 FEATURE REDUCTION

Feature reduction in the proposed system refers to selecting only the most important information from stored records while ignoring unnecessary or repeated data. Since the system manages structured pilot data rather than image features, reduction focuses on simplifying data for faster processing and display. The system extracts essential fields such as expiry dates, license type, medical status, and recent flight details, while less important information is not used in summary views.

4.4 RETRIEVAL-BASED CHATBOT

The retrieval-based chatbot plays an important role in the proposed aviation record management system. It allows users to quickly access their stored information using simple text queries instead of manually searching through different modules. When a user asks a question, the chatbot analyzes the keywords in the message and identifies the type of information requested, such as medical details, license information, profile data, or logbook entries. Based on this analysis, the system searches the database and retrieves the relevant records.

4.5 DATA PROCESSING AND RETRIEVAL

- **User Request Processing:** When a user submits data or asks a question, the system receives and processes the request through the backend server.
- **Authentication Check:** The system verifies the user's identity using secure login (JWT authentication) before allowing access.
- **Query Analysis:** For chatbot requests, keywords in the user's message are analyzed to identify the required information.
- **Database Search:** The system searches the MongoDB database for matching records based on the request.
- **Information Retrieval:** When users request data or interact with the chatbot, the system searches the database and retrieves the relevant information.

4.6 CONCLUSION OF SYSTEM DESIGN

Managing aviation records in the proposed system involves the following steps:

- **Data Preparation:** Pilot information such as profile details, medical certificates, licenses, and logbook entries are collected and organized in a structured format.
- **System Design:** The web application is built using the MEAN stack to handle user interactions, data processing, and storage efficiently.
- **Data Storage:** Instead of model training, the system stores validated records in the MongoDB database for future access and updates.
- **Authentication Process:** Secure login using JWT ensures that only authorized users can access their records.

CHAPTER 5

IMPLEMENTATION

AND

METHODOLOGY

CHAPTER 5

IMPLEMENTATION METHODOLOGY

5.1. DATA COLLECTION AND PREPROCESSING

Objective: Collect and prepare pilot records for storage and processing in the system.

- Collect user information such as profile details, medical certificates, licenses, and flight logbook entries through registration forms.
- Validate the entered data to ensure correctness and completeness.

Pseudocode:

```
request = get_user_request()
module = identify_module(request)
data = fetch_from_database(module)
processed_data = process(data)
display(processed_data)
```

5.2. FEATURE EXTRACTION USING KNN

Objective: Extract useful information from stored records and provide relevant results to users.

- Data is processed based on user requests or chatbot queries.

5.3 EXPIRY ANALYSIS AND STATUS MONITORING

Objective: Monitor document validity and identify records that are valid, expiring soon, or expired.

- It compares expiry dates with the current date to determine status.
- The system analyzes stored dates of medical certificates and licenses.

Pseudocode:

```
from tensorflow. keras. applications import
from tensorflow. keras. models import Sequential
from tensorflow. keras. layers import LSTM, Dense, Flatten,

Time Distributed import numpy as np
 Receive user query from web interface
 query = get_ user_ input ()
 Identify requested module (profile / medical / license / logbook)
 module = detect _ keywords(query)
 Fetch data from MongoDB database
 data = fetch _ records(module)
 Format the result for display
 response = format _ response(data)
 Send response to dashboard or chatbot
 display(response)
```

5.3. SYSTEM DEPLOYMENT AND OPERATION

Objective: Run the application and make it available for users.

- The system is deployed using Docker containers for easy setup and portability.
- Frontend, backend, and database services run together in the container environment.

Pseudocode:

from sklearn. metrics import classification_ report, roc_auc_score y_pred= model.predict(test_data) print(classification_report(test_labels, y_pred)) print("AUC-ROC Score:", roc_auc_score(test_labels, y_pred))

5.4. DEPLOYMENT

Objective:

Deploy the aviation record management system so users can access it in real time. The objective of deployment is to make the **Aviation Record Management System** accessible to users in real time so that pilots and administrators can efficiently manage aviation records such as medical certificates, licenses, personal profiles, and flight logbook details.

Pseudocode:

Start_ docker ()
connect_ database ()
request = receive_ user_ request ()
response = process_ request(request)
send_ response(response)

5.5. DEPLOYMENT

Objective:

Deploy the aviation record management system for real-time. The objective of deployment is to make the Aviation Record Management System available for real-time usage by pilots and administrators through a web-based platform. Deployment ensures that the developed system can be accessed, executed, and maintained efficiently in a production environment. The deployment process involves integrating the backend server, database, and user interface so that users can securely store and retrieve aviation records such as **pilot profiles, medical certificates, licenses, and flight logbook details**. The system is deployed using a **Flask web framework**, which enables the application to handle HTTP requests and provide responses through API endpoints.

Pseudocode:

```
Start_ docker ()

connect_ database ()

request = receive_ user_ request ()

response = process_ request(request)

send_ response(response) give the content
```

CHAPTER 6

RESULTS AND DISCUSSION

CHAPTER 6

RESULTS AND DISCUSSION

The system was designed to securely manage pilot records and provide quick access to information through a web interface and chatbot. The results obtained from testing demonstrate that the system performs efficiently under normal operating conditions.

6.1 EXPERIMENTAL SETUP

The experimental setup for evaluating the proposed system was designed to simulate real-time usage and assess its effectiveness across different scenarios. Testing focused on system functionality, response time, data accuracy, and security.

6.1.1 HARDWARE CONFIGURATION

- **Computer:** A standard desktop or laptop computer was used to develop, deploy, and test the system. The system included an Intel Core i5 processor (or equivalent), 8 GB RAM, and sufficient storage to run Docker containers smoothly.
- **Network Connectivity:** A stable internet connection was required to access the MongoDB Atlas cloud database and to test real-time system functionality.

6.1.2 SOFTWARE CONFIGURATION

- **Programming Language:** JavaScript / TypeScript
- **Operating System:** Windows 10 or abov

- **Frontend:** Angular — used to create the user interface.
- **Backend:** Node.js with Express.js — used to run the server.
- **Database:** MongoDB Atlas — used to store all pilot records.
- **Docker:** Used to deploy and run the application in containers.
- **Authentication:** JWT — used for secure login.

6.1.3 DATASET AND PREPROCESSING

The system uses pilot records instead of image datasets. The data consists of information entered by users and stored in the database:

- **Profile Data:** Personal details such as name, contact information, and email.
- **License Details:** Data related to pilot licenses and their validity.
- **Logbook Entries:** Flight details including hours flown and recent activities.

PREPROCESSING STEPS:

- **Data Validation:** Entered information is checked for correctness and completeness.
- **Duplicate Checking:** Existing records are verified to avoid repetition.
- **Security Processing:** Passwords are encrypted before storing in the database.

6.1.4 EXPERIMENTAL CONDITION

- **Different User Inputs:** The system was tested with various types of data entered by users, including profile details, medical records, license information, and logbook entries.

- **Multiple Access Scenarios:** Testing was performed for different operations such as registration, login, record updates, data retrieval, and chatbot queries.
- **Concurrent Usage:** The system was evaluated with multiple requests to ensure stable performance under normal usage conditions.

6.1.5 PERFORMANCE METRICS

- **System Reliability:** Evaluates whether the application runs without errors during normal usage.
- **Response Time:** Measures how quickly the system processes user requests and displays results.
- **Security:** Verifies that only authorized users can access and modify records.
- **Usability:** Assesses how easy the system is to use for managing records and interacting with the chatbot.

6.1.6 RECOGNTION RATE MEASUREMENT

The recognition rate was measured based on the percentage of correctly classified Skin images. An image was considered correctly classified if the KNN model accurately predicted the presence or absence of cancer. This metric helped assess the overall reliability of the system in detecting skin cancer under varying conditions.

6.2 IMPLEMENTATION

Pilot records such as medical certificates, licenses, and flight logbook details are entered and stored in the system to monitor pilot readiness and status.

6.2.1 Pilot Information Management System – Dashboard Interface

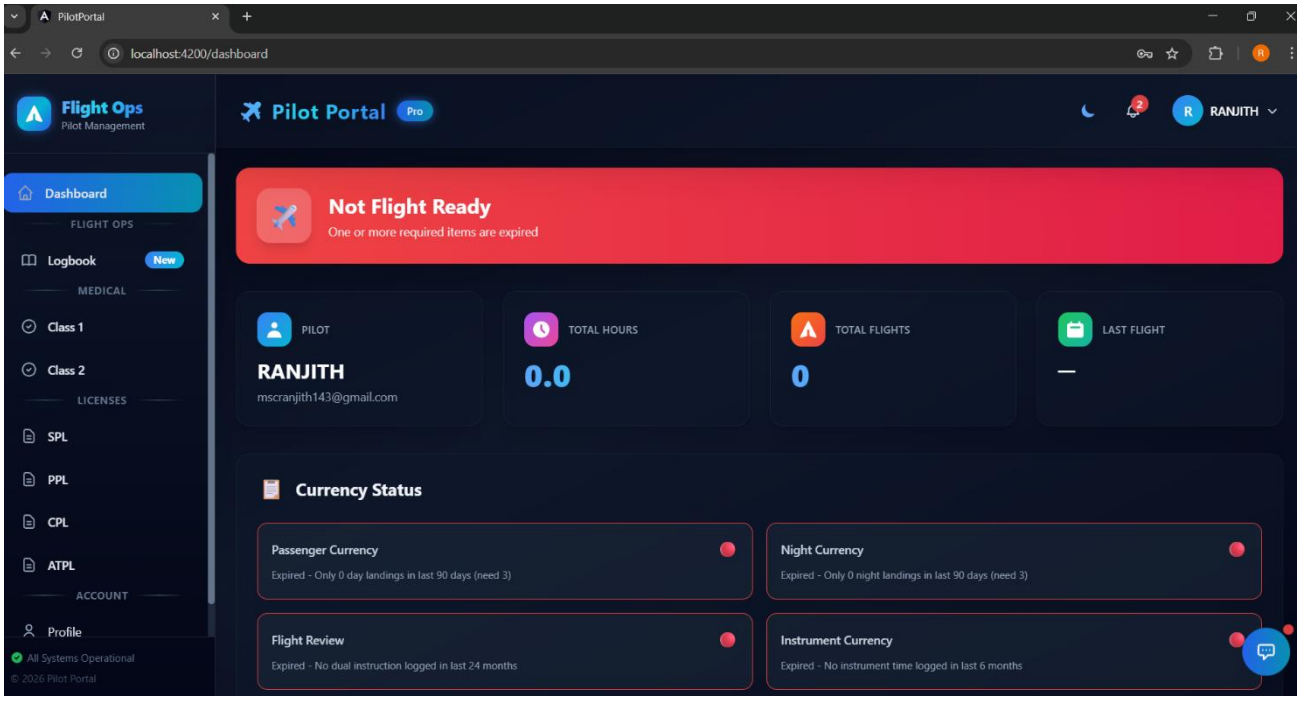

Fig 6.2.2 Dashboard Interface

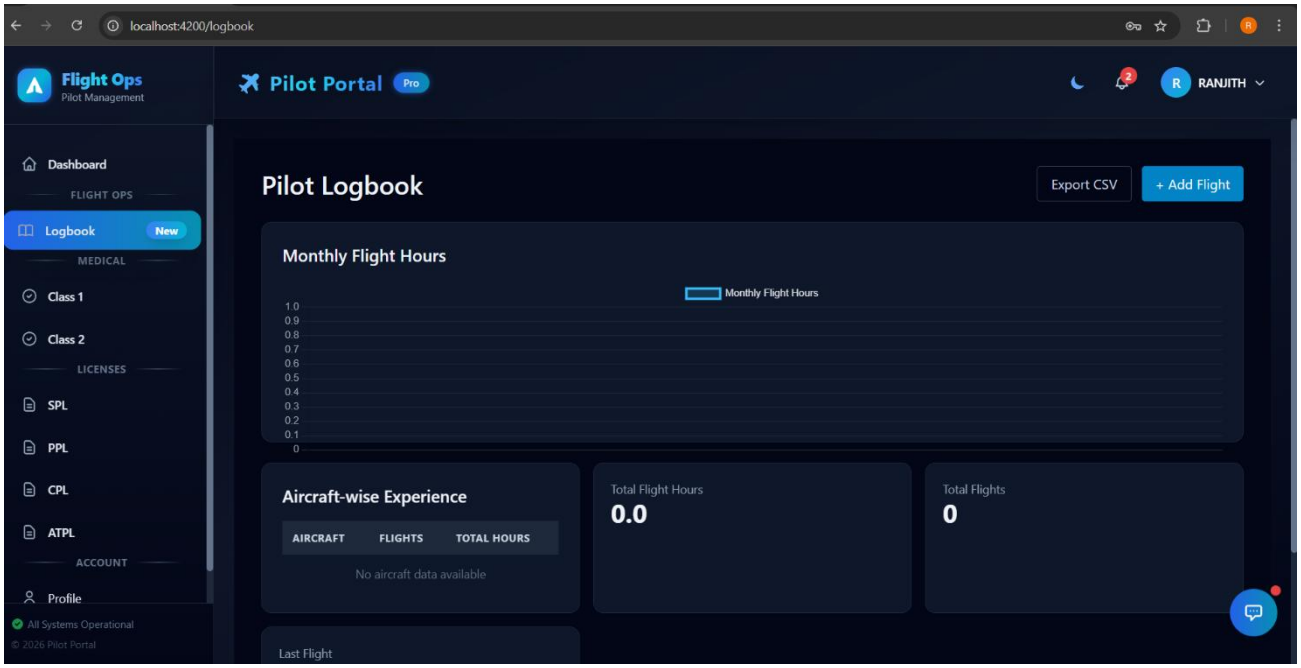

Fig 6.2.2 Flight Logbook Module

6.2.2 Medical Record Entry

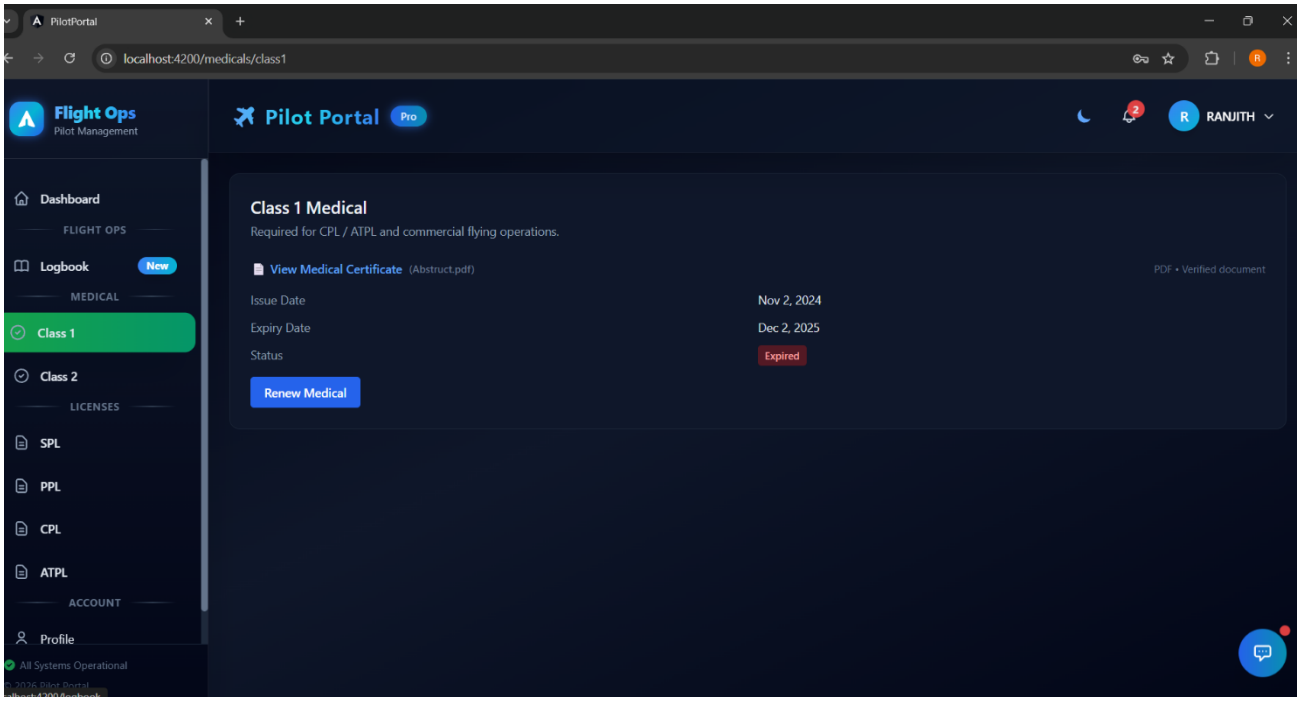

Fig 6.3 Medical inputs

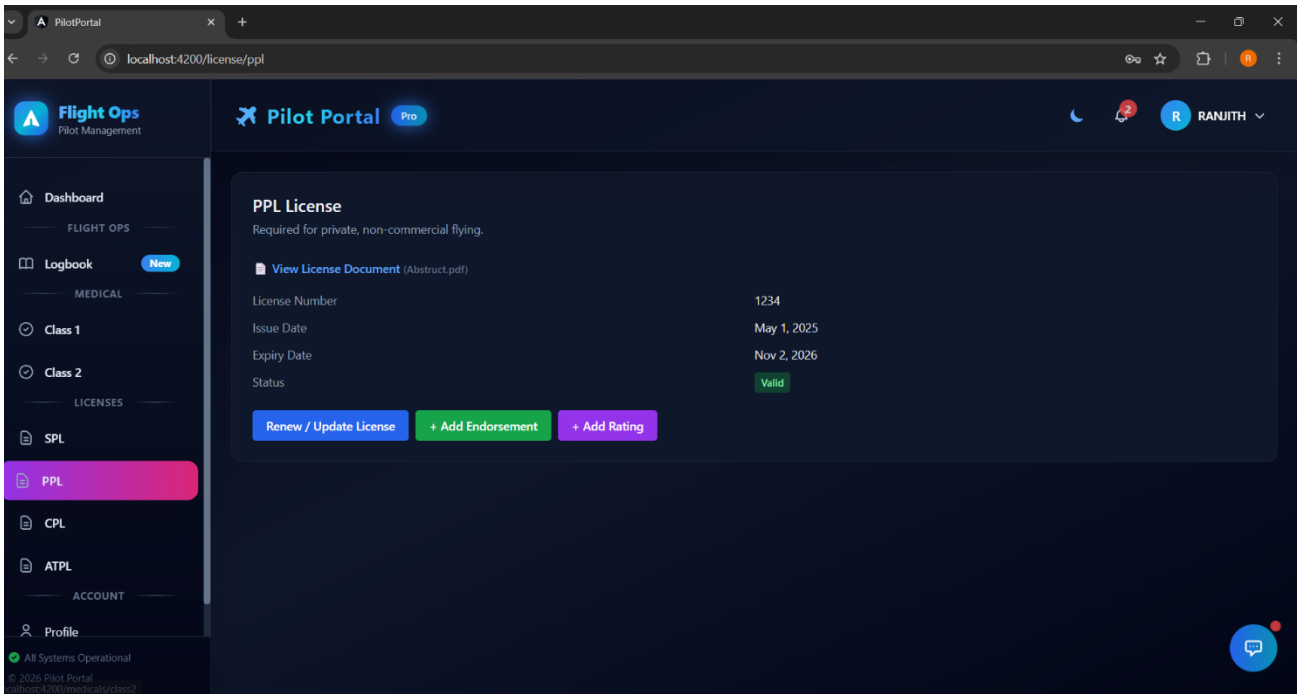

Fig 6.4 PPL License

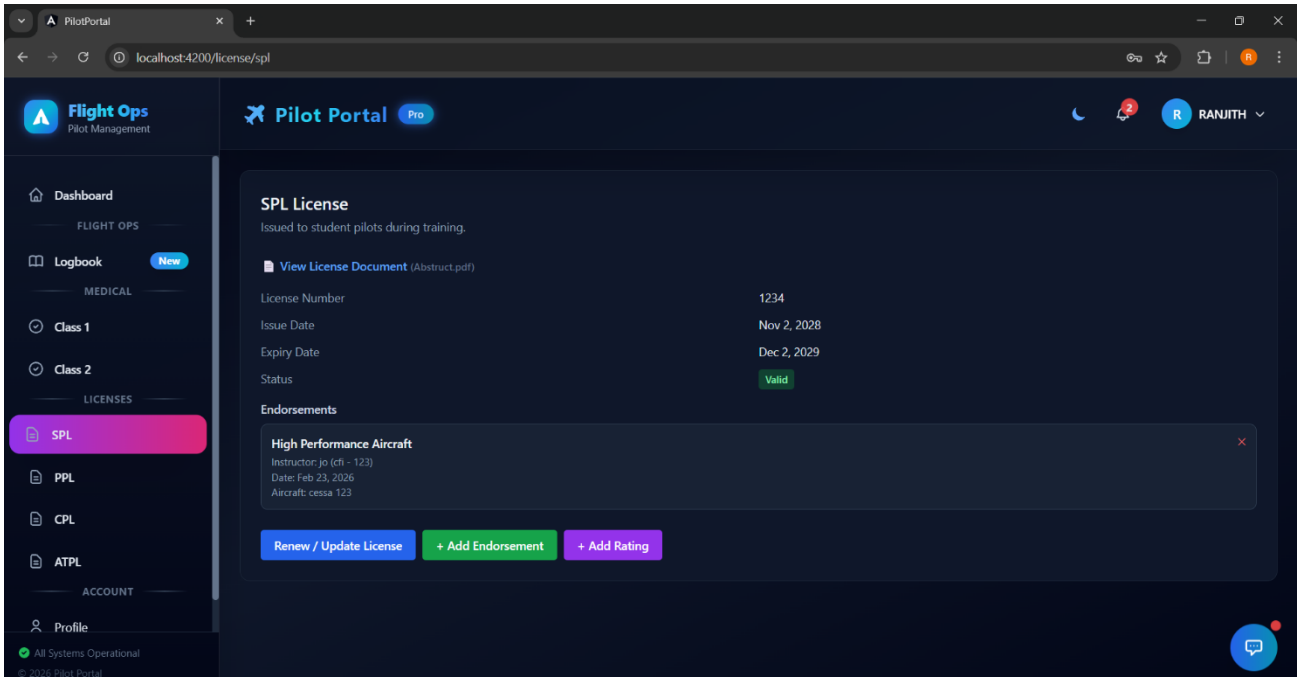

Fig 6.5 SPL License Renew/Update

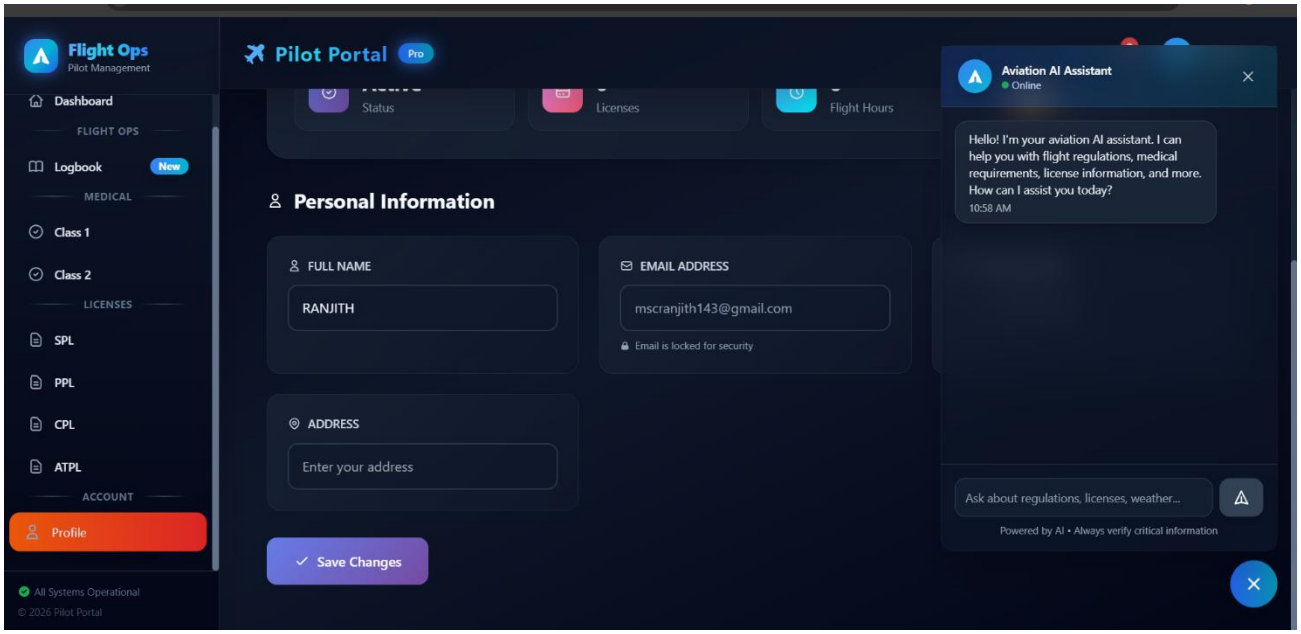

Fig 6.6 – Chat Bot

CHAPTER 7

CONCLUSION AND FUTURE WORKS

CHAPTER 7

CONCLUSION AND FUTURE WORK

This chapter presents the conclusion of the proposed system and discusses possible future enhancements that can further improve its functionality and usability.

7.1 CONCLUSION

In conclusion, the proposed Aviation Record Management System with Integrated Chatbot provides an efficient and secure solution for managing pilot records through a web-based platform. The system centralizes important information such as profile details, medical certificates, licenses, and flight logbook entries, reducing the need for manual record keeping.

7.2 FUTURE WORK

Future improvements to the system can focus on enhancing functionality, security, and user experience. Integration with official aviation databases or regulatory systems could enable automatic verification of licenses and medical certificates. Advanced chatbot capabilities using artificial intelligence and natural language processing can be implemented to support more complex conversations. Mobile application development would make the system accessible on smartphones and tablets. Additional features such as notification alerts for upcoming expirations, multi-user support for organizations, and role-based access control can further improve the system. Cloud deployment and advanced analytics for flight performance tracking can also be explored in future versions.

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CHAPTER 8

MAPPING

CHAPTER 8

MAPPING

8.1 PO MAPPING

The following outcomes are attained through the project titled "AVIATION RECORD MANAGEMENT SYSTEM".

S.NO	PO/ PSO	Engineering Graduate will be able to :	Program Outcome Relevance to the Project
1	PO1	**Engineering Knowledge** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization for the solution of complex engineering problems.	Uses computing and engineering concepts to design a secure system for managing aviation records and providing quick information through a chatbot.
2	PO2	**Problem analysis** Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	Analyzes issues in storing and managing pilot records and ensures secure and easy access to the data.

3	PO3	**Design & Development of solutions** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.	Designs a system to securely manage pilot records and provide easy access to important information in real-world aviation use.
4	PO4	**Conduct investigations of complex problems** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	Uses testing to ensure the system works correctly and provides accurate information to users..
5	PO5	**Modern tool usage** Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling to complex engineering activities, with an understanding of the limitations.	Uses MEAN stack tools and Docker to build and deploy the aviation record management system.

6	PO6	**The Engineer and Society** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice	Helps improve aviation safety by managing pilot records responsibly and efficiently.
7	PO7	**Environmental and sustainability** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development	Encourages digital systems that replace paper records and make aviation record management easier and more efficient.
8	PO8	**Ethics** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	Protects pilot information by ensuring data privacy and secure access in the system.
9	PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	Promotes teamwork during the development and documentation of the system.

10	PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Explains system performance clearly using reports, visuals, and presentations.
11	PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work.	Uses planning and management skills to complete the project and manage data effectively.
12	PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	Encourages learning new technologies to improve future aviation software systems.

		Program Specific Outcome (PSOs)	
	PSO1	**Professional Skills:** The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying complexity.	Uses software and data management methods to build an efficient aviation record system.
	PSO2	**Problem Solving Skills:** The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.	Follows practical development steps to create a reliable and easy-to-use aviation record system.

8.2 SDG MAPPING

The following Sustainable Development Goals (SDG) are mapped for the project titled "A Secure Web-Based Aviation Record Management System with Integrated Retrieval-Based Chatbot Using MEAN Stack".

S.NO	SDG	SDG	Mapped with the Project
1	SDG1	**No Poverty** End poverty in all its forms everywhere	
2	SDG2	**Not directly related** supports development indirectly through digital innovation.	

3	SDG3	**Good Health and well-being** Ensure healthy lives and promote well-being for all at all ages	Helps monitor pilots' medical records and certificate validity, supporting safety and well-being.
4	SDG4	**Quality Education** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	Provides practical learning in web technologies, cloud computing, and software engineering.
5	SDG5	**Gender Equality** Achieve gender equality and empower all women and girls	
6	SDG6	**Clean water and Sanitation** Ensure availability and sustainable management of water and sanitation for all	The system provides equal access to all users regardless of gender.
7	SDG7	**Affordable and Clean Energy** Ensure access to affordable, reliable, sustainable and modern energy for all	Digital systems reduce paper usage and resource consumption.
8	SDG8	**Decent work and Economic Growth** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	
9	SDG9	**Industry, Innovation and Infrastructure** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation	Promotes digital transformation in aviation using modern web technologies.
10	SDG10	**Reduced Inequalities** Reduce inequality within and among countries	

11	SDG11	**Sustainable Cities and Communities** Make cities and human settlements inclusive, safe, resilient and sustainable	Supports safer transportation systems through efficient aviation record management.
12	SDG12	**Responsible Consumption and production** Ensure sustainable consumption and production patterns	Encourages paperless operations through digital record keeping.
13	SDG13	**Climate Action** Take urgent action to combat climate change and its impacts	
14	SDG14	**Life below water** Conserve and sustainably use the oceans, seas and marine resources for sustainable development	
15	SDG15	**Life on Land** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	
16	SDG16	**Peace, Justice and Strong Institutions** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	Ensures secure data handling and transparency in record management.

CONFERENCE DETAILS

JOURNAL DETAILS

JOURNAL PUBLICATION

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Journal Published date: 01-01-2026

A SECURE WEB-BASED AVIATION RECORD MANAGEMENT SYSTEM WITH INTEGRATED RETRIEVAL-BASED CHATBOT USING MEAN STACK

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Keywords
Aviation Records, MEAN Stack, JWT Authentication, Chatbot, NLP, Retrieval Systems, MongoDB, Pilot License, Logbook Management, Web Application.

I. ABSTRACT

This paper presents the design and implementation of a secure, cloud-enabled Pilot Portal Application that digitalizes aviation record management, including medical certificates, flight logbooks, pilot licenses, and personal details. Developed using the MEAN stack (MongoDB, Express, Angular, Node.js), the system employs JSON Web Tokens (JWT) for authentication without relying on external IAM frameworks. A novel addition to the system is a retrieval-based chatbot that interacts with pilot data stored in the database, enabling users to obtain information using natural-language queries. The chatbot interprets user intent and retrieves corresponding entries from medical, license, profile, or logbook modules.

The system's architecture ensures modularity, scalability, and real-time dashboard generation, while the chatbot enhances accessibility and usability. This work demonstrates how conversational retrieval mechanisms can support aviation applications and reduce manual lookup time, providing an efficient digital ecosystem for pilots and training institutions.

II. INTRODUCTION

Pilots are required to maintain multiple regulatory and compliance-related documents, including medical certificates, logbook entries, and license records such as SPL, PPL, CPL, and ATPL. These records are traditionally maintained manually or stored across fragmented systems, creating challenges in accessibility, safety, and tracking expirations.

To address these challenges, we propose a web-based Pilot Portal Application built using modern full-stack web technologies. The system aims to centralize all pilot documentation, ensure secure access, and automate monitoring of expiry-sensitive items.

A key innovation introduced in this work is a retrieval-based chatbot, which allows pilots to query their data conversationally. Instead of navigating multiple screens, users can ask natural-language questions—such as "When does my Class 1 Medical expire?" or "Show my

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Authored by

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Kamaraj College of Engineering Technology.

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Department of Computer Science and Engineering

Vision of the institution

To make this Institution the unique of its kind in the field of Research and Development activities in this part of world.

Mission of the institution

To impart highly innovative and technical knowledge to the urban and unreachable rural student folks through "Total Quality Education".

Vision of the Department

To make the Department of Computer Science and Engineering the unique of its kind in the field of Research and Development activities in this part of world.

Mission of the Department

To impart highly innovative and technical knowledge to the urban and unreachable rural student folks in Computer Science and Engineering through "Total Quality Education".

Program Educational Objectives (PEO)

PEO 1: Apply the necessary mathematical tools and fundamental knowledge of computer science & engineering to solve variety of engineering problems.

PEO 2: Develop software-based solutions for real life problems and be leaders in their profession with social and ethical responsibilities.

PEO 3: Pursue life-long learning and research in selected fields of computer science & engineering and contribute to the growth of those fields and society at large.

Program Specific Outcomes (PSO)

PSO1: Professional Skills: The ability to understand, analyse and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying complexity.

PSO2: Problem - Solving Skills: The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.

Program Outcomes (PO)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Considers safety, legal, and social responsibilities while developing a secure aviation record management system.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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